electrical system, and when any such system is disturbed, electromagnetic waves are emitted. But the atomic nucleus, too, is an electrical system, as is shown by its charge, and so it can be reasonably expected that in conjunction with certain internal processes within the nucleus, the nucleus, too, will emit electromagnetic waves—gamma radiation.

In natural radioactivity, beta particles make an appearance always as carriers of a negative charge only, in other words, only as *negative* electrons. Let us jump ahead for a moment and mention that in the radiation of artificially produced radioactive substances *positively* charged electrons are observed also; they are particles having the same mass as the negative electrons, but each of them carries an elementary quantum of positive electricity (*Anderson*). To-day, these particles are known as *positrons*, and electrons are usually regarded merely as a negative variety of them. The question now arises, why these positrons were not observed a long time ago, and why they do not occur in the extranuclear structure of the atom. The answer to ·this question is based on experiments which have established that positrons are very short-lived. As soon as a positron approaches an electron—usually after a very brief interval of time—it combines with the latter, to form an electrically neutral structure; the product of this union is one or two gamma-ray photons, i.e. photons of an extremely short wavelength. These constitute what is known as *annihilation radiation.* Both the existence of positrons and of annihilation radiation were predicted by Dirac, and subsequently confirmed by experiments. The annihilation radiation has its counterpart, too; a photon entering the powerful field in the immediate vicinity of an atomic nucleus, can change into an electron and a positron. This *formation of a pair of charged particles* can be observed in the cloud chamber, and is shown in the photograph reproduced here as Figure 9. In the cloud chamber, a very strong magnetic field deflects the electrons to one side, and the positrons to the other, so that they describe circular tracks (Figure 9). In the upper half of our photograph, such a pair formation has just taken place, and the two tracks of the electron and positron thus formed can be clearly seen. Since this photograph is considerably enlarged, the individual cloud droplets along the tracks can also be recognized. Another electron, the track of which is slightly blurred, is visible, too. The rest of the droplets are due partly to impurities.